

Upcycling plastic waste into high value-added chemicals under mild conditions

Xiangyue Wei

College of Chemistry, Sichuan University

Qiang Zhang

College of Chemistry, Sichuan University

Chengfeng Shen

College of Chemistry, Sichuan University

Xu Zhao

College of Chemistry, Sichuan University

Fan Zhang

Sichuan University, College of Chemistry

Xuehui Liu

College of Architecture and Environment, Sichuan University

Gang Wu

College of Chemistry, Sichuan University

Shimei Xu (✉ xushimei@scu.edu.cn)

College of Chemistry, Sichuan University

Yu-Zhong Wang

State Key Laboratory of Polymer Materials Engineering, College of Chemistry, Sichuan University

Article

Keywords:

Posted Date: October 26th, 2022

DOI: <https://doi.org/10.21203/rs.3.rs-2183154/v1>

License: © ⓘ This work is licensed under a Creative Commons Attribution 4.0 International License.

[Read Full License](#)

Upcycling plastic waste into high value-added chemicals under mild conditions

Xiangyue Wei¹, Qiang Zhang¹, Chengfeng Shen¹, Xu Zhao¹, Fan Zhang¹, Xuehui Liu², Gang Wu¹, Shimei Xu^{1*}, Yu-Zhong Wang^{1*}

¹The Collaborative Innovation Center for Eco-Friendly and Fire-Safety Polymeric Materials (MoE), National Engineering Laboratory of Eco-Friendly Polymeric Materials (Sichuan), College of Chemistry, Sichuan University; Chengdu 610064, China.

²The Collaborative Innovation Center for Eco-Friendly and Fire-Safety Polymeric Materials (MoE), National Engineering Laboratory of Eco-Friendly Polymeric Materials (Sichuan), College of Architecture and Environment, Sichuan University; Chengdu 610064, China.

*Corresponding author. Email: xushimei@scu.edu.cn; yzwang@scu.edu.cn

Abstract: As the largest class of organic thermoplastic polymers, polyolefins generated huge amounts of plastic wastes, which cannot be chemically recycled into their monomers or high-value chemicals in a cost- or energy-efficient way. To address this challenging issue, we develop a novel strategy for upcycling polypropylene (PP) waste. An anionic surfactant-induced oxidative degradation method is developed to transform polypropylene into terminal ketone/carboxylic acid/olefin functionalized long-chain oligomers (mainly average C70), which are important feedstocks used in synthetic surfactants, poly- α -olefin lubricants, and polyesters. The reaction only needs to be heated at 60°C for 5 min, and the subsequent reaction is completed by oxidative self-heating. The fast self-heating temperature up to 280°C triggers the thermal cracking of PP. Coupling exothermic oxidative cracking with endothermic thermal cracking promotes oxygenation and dehydrogenation of PP. Controlling the amount of Sodium dodecyl sulfate could modulate the self-exothermic process, giving the products with different degrees of functionalization. The approach provides a mild, facile, and fast solution to obtain value-added chemicals from waste polyolefins.

1 Since 1950, global plastic production has accumulated to 8.3 billion tons, generating
2 about 6.3 billion tons of waste plastics. The number is expected to exceed 34 billion tons by
3 2050 with more than 12 billion tons of waste plastics. However, only less than 10% of these
4 waste plastics are recycled, and the rest are landfilled or incinerated, or directly discarded,
5 resulting in serious environmental pollution and resource waste¹. Chemical recycling has
6 emerged as a promising technology to address this issue by converting the waste plastics into
7 new products including small molecules, polymers, or materials. Polyolefin, such as
8 polyethylene (PE) and polypropylene (PP), account for more than 40% of plastics consumed².
9 The pyrolysis route is generally acknowledged as the easiest way to convert the polymers into
10 small molecules. Unfortunately, due to a high temperature above 400°C, the thermal pyrolysis
11 route is unselective and always produces a low-value mixture of gas hydrocarbons, fuel oil,
12 and solid char³. Although catalytic cracking⁴, hydrogenolysis⁵, and tandem catalytic cross
13 alkane metathesis⁶ were developed to improve the reaction selectivity, these methods only
14 provide shorter alkane products, which have a limited range of applications due to the lack of
15 functional groups. Besides, catalysts of specific molecular structure bring higher costs on most
16 occasions.

17 Producing functional groups is a promising but challenging method to realize upcycling
18 of polyolefin⁷. Typically, there are two strategies used: one is the functionalization of the C-H
19 bonds in polyolefins⁸ without cleavage of the polymer chain (Fig. 1a); the other is to introduce
20 functional groups during breaking the C-C bonds (Fig. 1b). For the first strategy, the C-H
21 bonds of PE can be converted to C-X (X = hydroxyl and carbonyl groups) catalyzed by
22 high-valent ruthenium-oxo species and the obtained functionalized PE is directly used as

adhesives for both metals and polyolefins⁹. In contrast, the introduction of functional groups during the degradation process is a more flexible and versatile solution to expand the application areas of the degradation products. Oxidative degradation is a simple way to introduce functional groups such as carbonyl and hydroxyl groups by oxygenation or alkenyl groups by dehydrogenation. However, in most cases, polyolefins are deeply oxidized to complex short-chain fatty acids (C1-C7) owing to an uncontrollable free radical reaction¹⁰⁻¹². To improve the selectivity, Chow et al. reported that pre-sulfonation of PE before oxidation which positioned for degradation to small molecular carboxylic acids by photo-Fenton¹³. However, this method improved the selectivity of butanedioic acid to a limited degree. Besides, sulfonyl chloride was used in the sulfonation which made the process complex and unfriendly. In general, a low carbon utilization rate of less than 40% is always accompanied during the oxidative degradation of polyolefin due to the production of a large amount of greenhouse gas CO₂. Very differently, Xie et al. proposed a new idea by completely mineralizing PE to CO₂¹⁴, and then CO₂ was immediately converted to high-value acetic acid in tandem by photocatalyst Nb₂O₅. However, the efficiency of CO₂ conversion to acetic acid was as low as 1.5 mg·g_{cat}⁻¹.

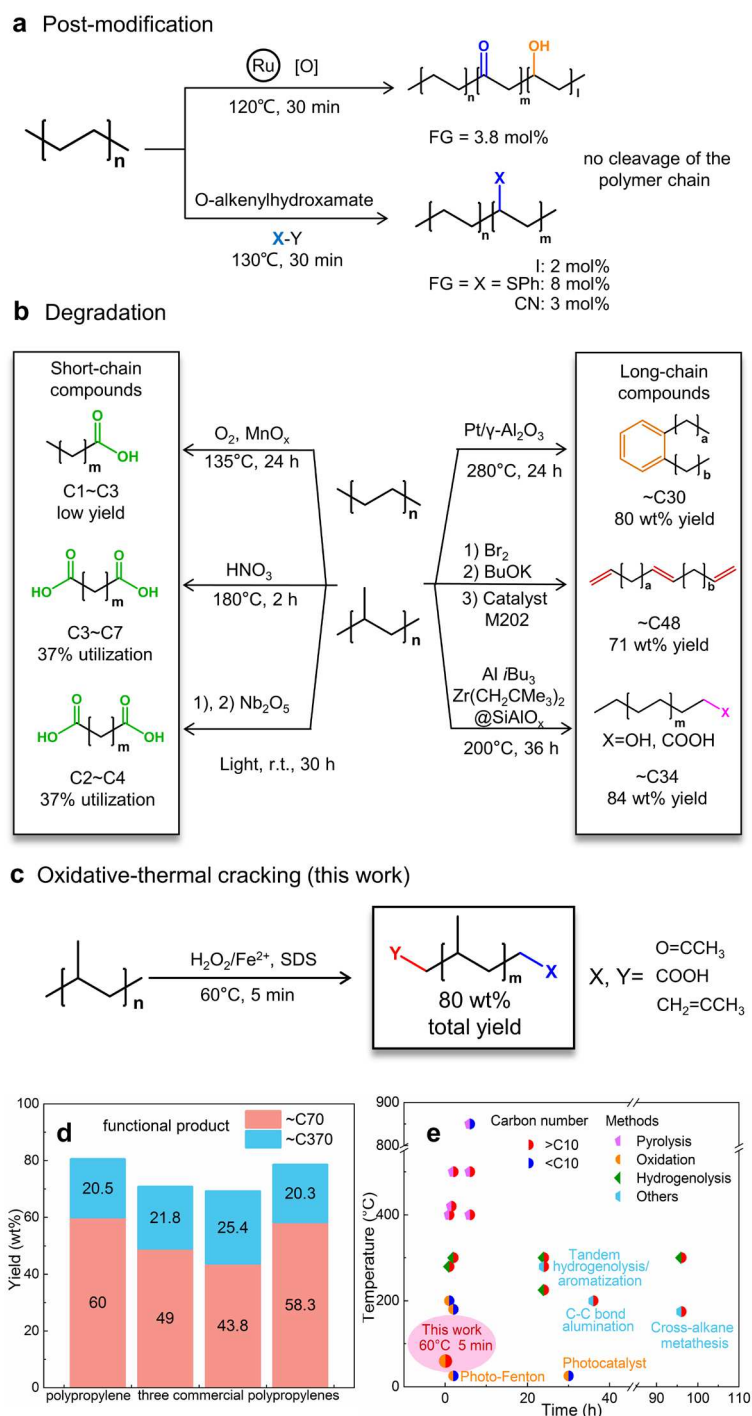


Fig. 1. Method for introducing functional groups into polyolefins. a, Free radical-mediated or organometallic-catalyzed C-H functionalization without polyolefin chain breakage; FG: functional group. **b,** Methods for degrading polyolefins into functional compounds; **c,** Introduction of methyl ketone, carboxyl and isopropenyl groups by self-exothermic oxidative degradation of PP. **d,** Functional products after degrading for 30 min at 80°C from different PP. **e,** Comparison of degradation effects between self-exothermic oxidative degradation and different degradation methods in references: pyrolysis^{4,15-17}, oxidation^{10,11,13,14}, hydrogenolysis^{5,18-21} and others^{6,22,23}.

Converting Polyolefins into functionalized compounds with long chains can improve the carbon utilization. Meanwhile, end-group functionalized long-chain compounds are valuable feedstocks and important structural motifs for a wide variety of high-value chemicals, polymers, and materials. Monofunctional long-chain compounds can be used in synthetic detergents, surfactants, poly- α -olefin lubricants²⁴, etc. Bifunctional ones can be used to synthesize monomers of condensed polymers²⁵ such as polyesters and polyamides, in which longer aliphatic segments endow polymers with new properties similar to polyolefins²⁶. The long-chain compounds are generally synthesized based on saponification of natural oils from biomass or oligomerization of olefin from petroleum. However, those traditional methods always require complex chemical synthesis^{27,28}, or biological ω -oxidation²⁹ to introduce functional groups at the α - and ω - positions. Chemical conversion of polyolefins would open up a new path for the preparation of the long-chain compounds. So far there is only limited success. Pyrolysis at a relatively low-temperature 370°C can convert PP into α,ω -divinyl-functionalized oligomers³⁰. Limited by the low efficiency of the reaction at a lower temperature, the molecular weight of the oligomers is high (average C450). The chain length of the oligomers can be decreased (average C48) via a tedious three-step process of bromination, dehydrobromination, and olefin metathesis reactions³¹. Recently, Zhang et.al. reported a tandem hydrogenolysis/aromatization process to convert the PE into long-chain alkylaromatics (average C24) by Pt/Al₂O₃ catalyst at 280°C for 24 h²². The alkylaromatics can be further sulfonated to make surfactants since it introduces the benzene ring in the degradation of polyolefins. By using C-C bond metalation strategies to produce a new C-O bond, Kunbur et al developed a new method to convert PE into long-chain fatty acids and alcohols (average C34) were obtained at 150°C for 12 h²³. These methods give new life to

polyolefins and open innovative ways for the upcycling of polyolefins. However, air-unstable catalyst precursors and noble metal catalysts complicated the manipulation. A simple and facile upcycling method of polyolefins into long-chain functional compounds is highly desired.

Results

Oxidative degradation of polypropylene and analysis of the products. Here, an anionic surfactant-induced oxidative degradation method is developed to transform PP into end-group functionalized long-chain chemicals with a yield up to 80% by simply using SDS and Fenton's reagent (Fig. 1c-e). To improve the mass and heat transfer in the degradation reaction, PP ($M_w=4.20\times 10^5$ g/mol, PDI=3.52) was first treated by a heating-cooling process with dodecane to obtain a loose white powder. The treated PP powder (0.4 g), SDS (0.16 g), $\text{FeSO}_4\cdot 7\text{H}_2\text{O}$ (7 mg), and H_2O_2 (3.6 mL, 30%) were then added into a hydrothermal synthesis kettle with a 40 mL PTFE liner. In a trial experiment, the degradation was carried out at 80°C for 30 min. After the reaction, the white PP powder was transformed into a homogeneous yellow paste. The three product fractions (solid, liquid, and gas) were further collected and analyzed. The solid phase is the main degradation product, which was separated into three substances as DPP-THF, DPP-Xylene and, Char by dissolving in THF and hot xylene sequentially, giving in 60.0 wt%, 20.5 wt%, and 1.3 wt% yields (based on initial PP mass), respectively (Fig. 2b). The IR analysis (Extended Data Fig. 1) showed that the solid product contained mainly ketone carbonyl group (1715 cm^{-1}), carboxyl group (1765 cm^{-1} and 3440 cm^{-1}) and alkenyl group (1640 cm^{-1}), and the relative intensity of these functional groups in DPP-THF was higher than the one in DPP-Xylene. Char is a black powder with almost no C-C

and C-H absorption, which is a carbonization product. Mw of DPP-THF is 1560 g/mol (PDI=1.95), while Mw of DPP-Xylene is 7707 g/mol (PDI=1.98). The liquid phase consists of C1~C8 fatty acids and ketones, of which the major specie is succinic acid (4.1 wt%, based on initial PP and SDS mass) (Extended Data Fig. 2). The total organic carbon (TOC) test showed that only 10.2% of the carbon in the raw material is converted into the aqueous phase product. The gas products were determined to be only CO₂ and CO without alkanes or olefins by GC-TCD analysis (Extended Data Fig. 3).

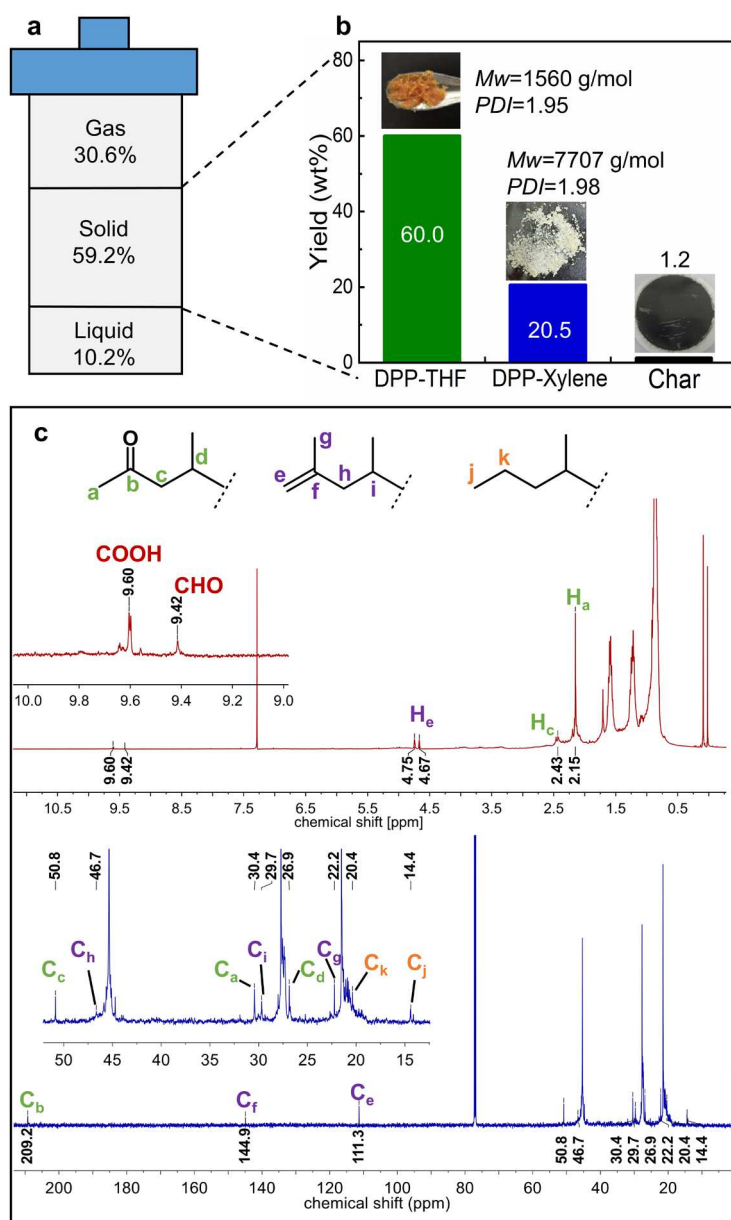


Fig. 2. Oxidative degradation of PP at 80°C for 30 min. **a**, Carbon balance of oxidative degradation based on PP and SDS. **b**, Yield base on PP and appearance of solid products. **c**, ^1H and ^{13}C NMR spectrum of DPP-THF in CDCl_3 .

The study of the ^1H and ^{13}C Nuclear magnetic resonance (NMR) spectrum provide more information about the individual molecular component in DPP-THF (Fig. 2c) and DPP-Xylene fraction. One type of end-functional group was determined as ketone carbonyl judged by the intense signal at 2.1 ppm in the ^1H NMR which assigns to direct association of methyl group with a carbonyl group ($\text{CH}_3\text{-C(O)-CH}_2\text{-}$) as well as a signal at 210 ppm in the ^{13}C NMR for ketone ($\text{CH}_3\text{-C(O)-CH}_2\text{-}$). Besides, the resonances at 9.6 ppm and 9.4 ppm in the ^1H NMR indicate that the DPP-THF fraction contains carboxyl and aldehyde groups respectively. There are no characteristic resonances for carboxyl and aldehyde groups in ^{13}C NMR (160-200 ppm), probably due to the weak intensity of the carbon spectrum signal itself. ^1H NMR spectrum contains the couple resonance signals at 4.6 and 4.7 ppm, and the ^{13}C NMR spectrum at 111 and 144 ppm indicate that they are characteristic signals of terminal alkenes (isopropenyl, $\text{CH}_2\text{=C(CH}_3\text{)-CH}_2\text{-}$). Proton associated with the direct bond to the alkyl and carbonyl groups resonate in the region 2.2-2.5 ppm. Furthermore, in the ^{13}C NMR spectrum of DPP-THF, the signals at 11.4 ppm, 22.2 ppm, and 26.9 ppm were assigned to terminal methyl ($\text{-CH}_2\text{-CH}_3$), methyl in terminal alkene ($\text{-C(CH}_3\text{)=CH}_2$) and methyl in methyl ketone ($\text{-CH}_2\text{-C(O)-CH}_3$), respectively^{25,30}. We conclude that both ketones and alkenes in DPP-THF bond to terminal groups, while carboxyl and aldehyde groups are located in terminal positions. Our main product, DPP-THF, are terminal functionalized long-chain aliphatic hydrocarbons, which are interesting as higher added value chemical products with wide applications.

The amount of methyl ketone and isopropenyl components in DPP-THF were quantified by ^1H NMR spectroscopy. At the ratio of SDS to PP of 0.40, the contents of methyl ketone and isopropenyl in DPP-THF were 1.29 mmol/g and 0.56 mmol/g, respectively (Extended Data Table 1, entry 4). The carboxyl content was determined as 1.26 mmol/g by acid-base titration. The oxygen content in the carboxyl and methyl ketone in DPP-THF was 2.07% and 4.03%, respectively (Extended Data Table 2, entry 3). The total amount of the two was similar to the total oxygen content (6.51%) measured by organic elemental analysis, indicating that most of the oxygen-containing groups in the product are relatively more stable methyl ketone and carboxyl groups rather than aldehyde groups.

There are characteristic signals of isopropenyl and methyl ketone in the ^1H NMR spectrum of DPP-Xylene in ^1H NMR spectrum (Extended Data Fig. 4) which is highly similar to that of DPP-THF. Considering the higher molecular weight and the same functional group species of DPP-Xylene, we assume that DPP-Xylene is an intermediate for the conversion of PP to DPP-THF.

The results revealed that PP was upgraded to valuable terminal carboxyl, methyl ketone and isopropenyl functionalized oligomers by simple Fenton oxidation, which has potentially widespread application as synthetic polymers, lubricants, paints and, adhesives.

Reasons for fast self-exothermic oxidation. We further explored the effect of heating temperature and time on the reaction. Even the reaction was heated at 60°C for only 5 min followed by removing the heat source and standing at room temperature for another 25 min, PP was still degraded into similar products mentioned above. As well known, PP is composed of stable C-C bonds and C-H bonds with large bond energy, and harsh conditions such as high

temperature ($> 450^{\circ}\text{C}$) are usually required to cleave the inert bonds^{3,15}. However, in our oxidation system, C-C can be broken at a low heating temperature within several minutes. So we speculate that there is additional energy generation during the reaction process. As expected, a sharp temperature rise was observed during the oxidative degradation reaction. It is clearly seen in Fig. 3a that the internal temperature of the reaction rises rapidly from 40°C to 266°C in 26 s. The exothermic oxidation reaction provides additional energy to the system, and PP is fast degraded by self-exothermic oxidation. The self-exothermic phenomenon originates from the oxidative decomposition of organic matter, and the contribution of PP and SDS to the self-exothermic process was investigated. A rapid rise of the internal temperature from 40°C to 165°C was observed but in a longer time (46 s) when SDS was degraded in the absence of PP (Fig. 3b). Under different dosages of SDS, the internal peak temperature of PP degradation is always higher than that of SDS decomposition alone, but the time to reach max temperature is similar in the two cases (Fig. 3d). Meanwhile, we carried out the decomposition of Fenton's reagent as a control experiment, finding that the internal temperature rose slowly, and the highest was only 73°C (Fig. 3c). It can be concluded that the oxidative degradation of PP was induced by the oxidative decomposition of SDS.

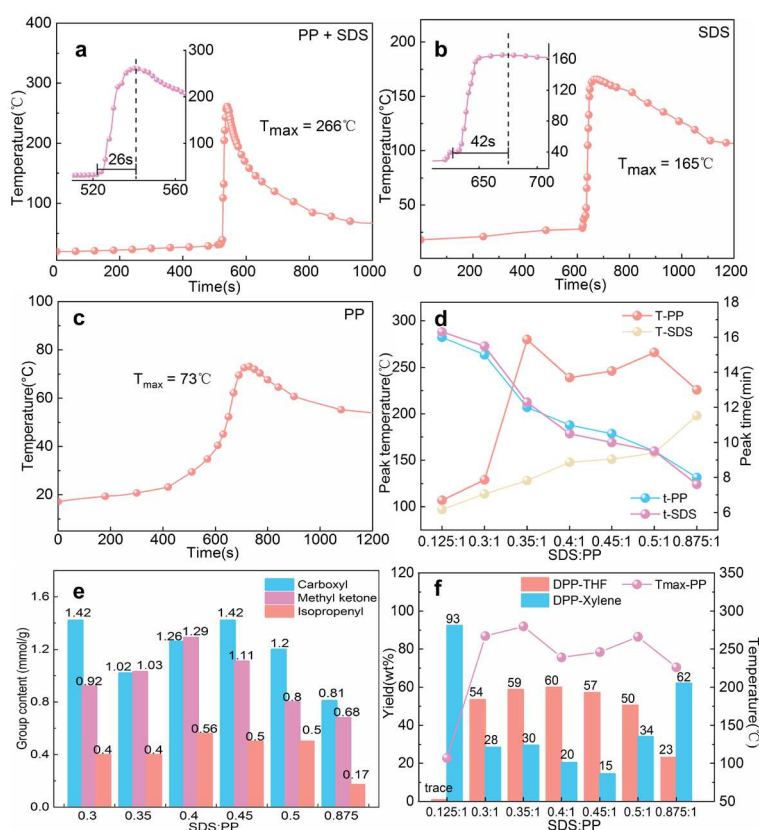


Fig. 3. Effects of SDS on the self-exothermic oxidative degradation of PP. **a**, SDS-assisted oxidative degradation of PP at SDS:PP=0.5. **b**, The oxidative degradation of SDS without PP. **c**, The oxidative degradation of PP without SDS. **d**, The relationship of internal peak temperature and time in degradation reaction of PP varying the weight ratio of SDS to PP. **e**, Effect of weight ratio of SDS to PP on functional group content in DPP-THF. **f**, Effect of internal peak temperature on yield of DPP-THF and DPP-Xylene.

The amount of SDS affects the internal temperature and thus the products distribution (Fig. 3f). In the ratio range of SDS to PP of 0.125:1~0.5:1, the internal peak temperature has experienced a process of increasing, decreasing, and then increasing. When the internal peak temperature approaches the valley bottom, the yield of DPP-THF increases to the maximum value of 60.0 wt% while that of DPP-Xylene is down to 20.5 wt%. It further confirmed that more DPP-Xylene was converted into DPP-THF. This increased cracking reaction is strongly endothermic, causing the internal peak temperature to decrease. At the same time, the content of isopropenyl in the main product DPP-THF is rising (Fig. 3e) when the internal peak temperature decreases. Combining the changes in internal peak temperature and isopropenyl

content, we speculate that thermal cracking takes place in the degradation of PP. In addition, the content of methyl ketones and carboxyl groups, which are from oxidative cracking, increases with the increase in isopropenyl content. This shows that oxidative cracking and thermal cracking promote each other. SDS regulates the degree of oxidative cracking of PP by consuming the oxidant H_2O_2 , which further affects the reaction temperature and the degree of thermal cracking.

In addition to the internal temperature effect, SDS acts other key roles. As an anionic surfactant, SDS is inclined to be oriented on the PP/water interface with the sulfonate head group toward the water. So the interface property of PP is changed. On one hand, the interface becomes more wetting to water; on the other hand, it allows to adsorb Fe^{2+} in Fenton's reagent with no need of additionally grafting negatively charged groups^{13,32}. These Fe^{2+} absorbed on the PP interface increase the concentration of hydroxyl radicals while shorten the effective diffusion distance of the radicals from the bulk solution to the PP interface. It significantly promotes the degradation of PP. When SDS was replaced by cationic surfactants, PP is hardly degraded due to the lack of the adsorption of Fe^{2+} on the interface (Extended Data Fig. 5).

Mechanism of producing end-group functionalized long-chain chemicals. Combining the structure changes of intermediates and products, we propose the main cracking mechanism in this process (Fig. 4). Firstly, the solvent treatment of dodecane destroys the crystalline regions of PP and facilitates the mass and heat transfer during the degradation reaction. SDS was then adsorbed on the PP/water interface and attracts Fe^{2+} on the interface from the solution through electrostatic interaction. The attracted Fe^{2+} facilitated the generation of

hydroxyl radicals from H_2O_2 , and the high concentration of hydroxyl radicals on the interface significantly improves the initiation reaction efficiency to convert PP to tertiary carbon radicals, which were further transformed into alkoxy radicals. Next, β -scission from the alkoxy radicals results in C-C bond cleavage to give a terminal methyl ketone³³. The alkoxy radical of the secondary carbon generates an aldehyde group through β -scission, and the aldehyde group is further oxidized to generate a carboxyl group³⁴. Tertiary carbon radicals can also be generated via homogeneous C-C bond cleavage and intermolecular hydrogen transfer. β -scission of the tertiary carbon radicals results in a terminal olefin³⁰. The coupling of the exothermic and endothermic reactions leads to the conversion of PP to methyl ketone, carboxyl, and isopropenyl-functionalized oligomers. It was worthy to be mentioned that SDS was oxidized into small molecular acids, which mainly exist in the liquid phase.

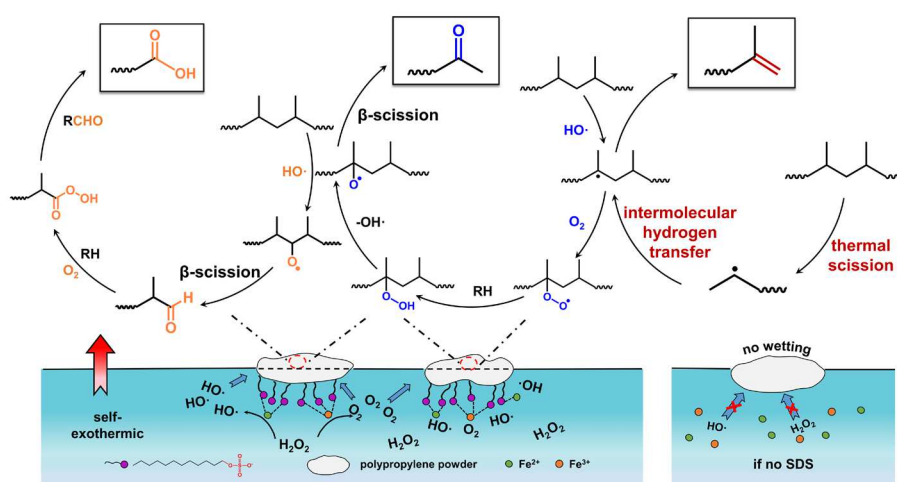


Fig. 4. Mechanisms for obtaining methyl ketone and isopropenyl from PP by tandem oxidative and thermal cracking.

Thermodynamic analysis of the oxidative cracking and thermal cracking was performed using the Benson group contribution method (Extended Data Table 3). The oxidative cracking of PP to form methyl ketone is exothermic ($\Delta H^\circ_1 = -251.9 \text{ kJ/mol}$) and thermodynamically spontaneous ($\Delta G^\circ_1 = -296.2 \text{ kJ/mol}$). It was similar to the one to form carboxyl ($\Delta H^\circ_2 = -594.8$

kJ/mol, $\Delta G^\circ_2 = -620.3$ kJ/mol). Differently, the thermodynamic data of the thermal cracking of PP to form isopropenyl groups is $\Delta H^\circ_3 = 86.1$ kJ/mol and $\Delta G^\circ_3 = 36.9$ kJ/mol respectively, indicating that it is difficult to proceed spontaneously at low temperatures. However, the ΔG° for the overall reaction (oxidative - thermal cracking) is -259.3 kJ/mol (coupling isopropenyl formation with methyl ketone formation) or -583.4 kJ/mol (coupling isopropenyl formation with carboxyl formation), making the thermal cracking reaction thermodynamically favorable.

In our degradation system, the temperature of thermal cracking is significantly lower (280°C) compared with one of conventional pyrolysis (>350°C) due to the exothermic oxidation reaction. Under the air atmosphere, the initial decomposition temperature and maximum decomposition temperature of PP are 287°C and 392°C respectively (Extended Data Fig. 6b), which are significantly lower than those (415°C and 458°C) under the N₂ atmosphere (Extended Data Fig. 6a). In addition, we heated PP to 390°C and 410°C under the N₂ atmosphere respectively, and then switched to air for isothermal decomposition for 30 min, and monitored the enthalpy during the decomposition process by DSC. It was found that the enthalpy at 410°C (Extended Data Fig. 6d) is lower than 390°C (Extended Data Fig. 6c) due to more thermal cracking reactions taking place, absorbing the heat released by the oxidative decomposition. It is worth noting that the thermal cracking reaction of PP does not occur at 410°C under the N₂ atmosphere, which indicates that the oxidative degradation of PP reduces the thermal cracking temperature of PP. The key to the generation of isopropenyl in thermal cracking of PP is to generate tertiary carbon alkyl radicals. Therefore, we compared the activation energy (E_a) of the elementary reactions that generate tertiary carbon alkyl radicals in the traditional thermal cracking and oxidative-thermal cracking processes (Extended Data

Fig. 7). In traditional thermal cracking, the E_a of elementary reactions are greater than 350 kJ/mol, while the E_a are less than 150 kJ/mol in the oxidative-thermal cracking. It can be explained that there are more ways to generate tertiary carbon radicals due to the addition of reactive oxygen species (ROS). Therefore, the temperature required for the thermal cracking is significantly lower than that of conventional thermal cracking.

Degradation of commercial polypropylene waste. We also performed several commercial PP degradation reactions to investigate the adaptability of the method (Fig. 5). Different from PP resins, commercial PP always contain additives such as antioxidants and light stabilizers. The commercial PP showed similar degradation behaviors with PP resins. PP food packing box ($M_w=1.87\times10^5$ g/mol, PDI 3.55) yielded 43 wt% DPP-THF, while PP centrifuge tubes ($M_w=3.07\times10^5$ g/mol, PDI 3.61) with ink printed on the surface were also successfully converted to DPP-THF with a yield of 58.3 wt%. We also challenged the more difficult PP bottle containing pigments ($M_w=3.42\times10^5$ g/mol, PDI 5.08) and a yield of 49 wt% of DPP-THF was achieved as well. The three commercial PP plastics gave methyl ketone, carboxyl, and isopropenyl -functionalized oligomers similar to the one from the PP resins degradation.

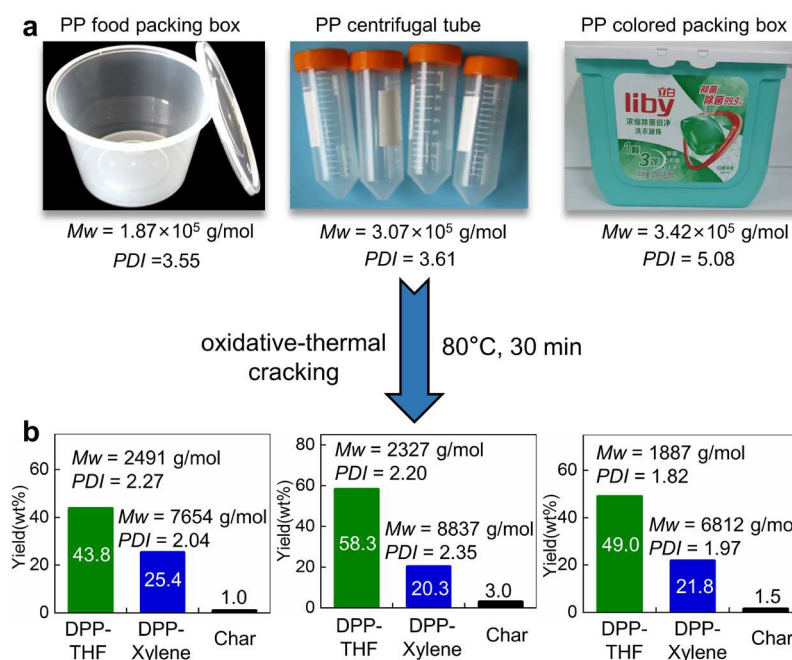


Fig. 5. Degradation of commercial PP into functional compounds. **a**, three commercial PP: PP food packing box, PP centrifugal tube and PP colored packing box. **b**, yield of solid products through oxidative-thermal cracking at 80°C for 30 min.

Discussion

In conclusion, methyl ketone, carboxyl, and isopropenyl functional groups were efficiently introduced into the degradation products by fast self-exothermic oxidation of PP. The directional aggregation of SDS and adsorption toward Fe^{2+} at the interface of PP/water renders the high-efficiency free radical reaction. Rapid oxidation reaction avoids the big reduction of chain length and thus long chain functional oligomers are obtained. Controlling the amount of SDS can modulate the self-exothermic process, giving the products with different functionalization. The end group modified oligomers have a high value for serving as feedstock in the synthetic polymers, lubricants, polyesters, etc. We propose an oxidative-thermal cracking mechanism where the combination of exothermic and heat absorption promotes the degradation of PP. The idea to convert waste plastics into valuable

long-chain chemicals has the great potential to expand applications for plastics chemical upcycling.

Method

Materials. Polypropylene pellets (PP-T30S, $M_w=4.20 \times 10^5$ g/mol, PDI 3.52) were purchased from China National Petroleum Corporation. $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ were purchased from Jinshan Chemical Test. Sodium dodecyl sulfate (SDS, $\geq 99.0\%$) were purchased from adamas-beta. Cetyltrimethylammonium Bromide (CTAB, $\geq 99.0\%$) and 30% H_2O_2 (AR) were purchased from Chron Chemicals. Xylene (AR) and THF (AR) were purchased from Tianjin Zhiyuan Reagent Co. Ltd.

Oxidative Degradation of PP. PP (3 g) was added to 10 mL of dodecane and heated at 160°C for 30 min under stirring. Afterwards, the PP was precipitated from the solution with temperature cooling. The treated PP was soaked with ethanol for 12 h to wash away dodecane, and then dried at 80°C followed by crushing into powder (80 to 100 mesh) for degradation experiment.

Degradation experiments were performed in a hydrothermal synthesis kettle with a 40 mL PTFE liner. Typically, 0.16 g SDS and 0.40 g PP powder (SDS: PP=0.4:1(w/w)) were added to 3.6 mL of 30% H_2O_2 under magnetically stirring. $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ (7 mg) was added after PP was fully wetted and dispersed well. The hydrothermal synthesis kettle was quickly sealed and transferred to an oil bath at 80°C for 30 min with a stirring speed of 600 rpm. After the reaction was completed, the hydrothermal synthesis kettle was placed in cold water to quench the reaction. After cooling down to room temperature, the reaction gases were collected using a gas bag while the solids and liquids were separated by filtration.

1 The solid products are dissolved in THF and the insoluble fraction was filtrated off. The
2 solvent was evaporated by a rotary evaporator to obtain a yellow paste (the main solid
3 product DPP-THF). The fraction insoluble in THF was transferred to xylene and the mixture
4 was heated at 100°C for 30 min followed by immediate filtration. The insoluble residue was a
5 small amount of black powder (Char), while the solution fraction was evaporated to obtain a
6 light-yellow powder (DPP-Xylene).

7 After degradation, the solution was highly acidic (pH 1). To characterize and quantify
8 the acid, a methyl ester derivatization reaction was carried out using 4 mL of the degradation
9 solution and 15 mL of 20% H₂SO₄-CH₃OH at 60°C for 6 h. The derivatized degradation
10 solution was extracted with dichloromethane for GC-FID and GC-MS analysis. The succinic
11 acid in the degradation solution was quantified by the standard curve of dimethyl succinate.
12 The total amount of organic matter in the degradation solution was determined by TOC
13 analysis. In addition to the degradation products of PP, the decomposition products of SDS
14 are also mainly included in the degradation solution. A certain amount of the degradation
15 solution was extracted with DCM to remove most of the organic components. The
16 concentration of SO₄²⁻ in the extracted degradation solution was measured by anion
17 chromatography. After deducting SO₄²⁻ in the raw material, the decomposition rate of SDS
18 was calculated according to the remaining SO₄²⁻ concentration. Next, the amount of
19 undecomposed SDS in the degradation solution was measured by methylene blue
20 spectrophotometry.

21 The reaction gases composition was determined by GC-TCD analysis. The gaseous
22 products are mainly CO₂ and a small amount of CO.

Adsorption of PP for Fe²⁺ in the reaction solution. The adsorption capacity of PP for Fe²⁺ in SDS or CTAB solution was studied. 0.16 g of SDS or CTAB were added to 5 mL H₂O, then 0.4 g of PP powder, and 7 mg of FeSO₄·7H₂O were added. The mixture was allowed to stand for 10 min at room temperature until the adsorption equilibrium. The Fe²⁺ concentration in the supernatant was measured by phenanthroline spectrophotometry, and the adsorption amount of PP to Fe²⁺ was calculated.

Monitoring of reaction internal temperature. The degradation reactions were carried out in a reactor with the temperature sleeve. In order to prevent the metal sleeve from affecting the degradation reaction, the temperature measuring point was designed above the reaction liquid level. The temperature of the gas in the reactor was measured, and the test temperature would be lower than the actual reaction temperature. The internal temperature of the reaction for different ratios of SDS to PP was recorded, and the curve of the internal temperature with time was plotted.

Degradation of commercial PP. Three commercial PPs (PP food packing box, PP centrifugal tube, and PP colored packing box) were pulverized into powder in the same way as PP-T30S. The degradation of commercial PP was carried out according to the optimal conditions for PP-T30S degradation. 0.4 g commercial PP powder and 0.16 g SDS (SDS:PP=0.4:1(w/w)) were added to 3.6 mL of 30% H₂O₂ under magnetically stirring. FeSO₄·7H₂O (7 mg) was added after PP was wetted and stirred well. The hydrothermal synthesis kettle was quickly sealed and transferred to an oil bath at 80°C for 30 min with stirring speed of 600 rpm. After the reaction was completed, the hydrothermal synthesis kettle was placed in cold water to quench the reaction. The products are separated in the same way

as before.

Yield and carbon balance. The yields of three solid products including DPP-THF ($Yield_{DPP-THF}$), DPP-Xylene, DPP-Xylene ($Yield_{DPP-Xylene}$) and Char and Char ($Yield_{Char}$) were calculated using eq. S1 respectively.

$$Yield_{solid\ product}(wt\%) = \frac{mass\ of\ solid\ product}{mass\ of\ initial\ PP} \times 100\% \quad S1$$

The yield of major liquid product succinic acid was calculated using eq. S2

$$Yield_{succinic\ acid}(wt\%) = \frac{mass\ of\ succinic\ acid}{mass\ of\ PP + mass\ of\ SDS} \times 100\% \quad S2$$

Reaction carbon balance. The carbon in the reaction feedstock SPP and SDS was transferred to the gaseous, solid and liquid products, where the percentage of carbon in the gaseous products was calculated by eq. S3.

$$C_{gas}(\%) = 1 - C_{solid} - C_{liquid} \quad S3$$

C_{solid} in the equation was obtained by organic elemental analysis of the solid product, and C_{liquid} was obtained by TOC analysis of the degradation solution.

Solid product characterization analysis. Content of methyl ketone and isopropenyl groups. 1H NMR was performed on a mixture of DPP-THF and 2.0 μL DCE in $CDCl_3$. The contents of methyl ketone and isopropenyl groups are calculated using eq. S5-S7.

$$mol\ of\ isopropenyl\ (mmol) = \frac{integration\ of\ isopropenyl}{integration\ of\ DCE} \times mol\ of\ DCE \times \frac{4}{2} \quad S4$$

$$mol\ of\ methyl\ ketone\ (mmol) = \frac{integration\ of\ Methyl\ ketone}{integration\ of\ DCE} \times mol\ of\ DCE \times \frac{4}{3} \quad S5$$

$$Content\ of\ functional\ group\ (mmol/g) = \frac{mol\ of\ function\ group}{mass\ of\ DPP-THF} \quad S6$$

Content of carboxyl group. A 10 g/L phenolphthalein ethanol solution and a 0.02 mol/L KOH ethanol standard solution were prepared. Blank titration: 20 mL of THF was added to a 250 mL Erlenmeyer flask. Add 2~3 drops of phenolphthalein solution as an indicator. The

titration was performed with KOH standard solution, and the appearance of a persistent pink color that did not fade for 30 s was considered as the endpoint of the titration. Then, the volume of KOH standard solution consumed was recorded. Sample titration: 20 mg of DPP-THF ($W_{\text{DPP-THF}}$) was weighed and dissolved in 20 mL of THF. 2~3 drops of phenolphthalein solution were added to the sample solution. The sample solution was titrated with KOH solution until the solution turned pink and kept the color for 30 s without fading. The volume of KOH solution consumed was recorded as $V_{\text{DPP-THF}}$, and the carboxylic acid content in DPP-THF was calculated after deducting the blank volume V_{blank} (eq. S7).

$$\text{Carboxyl content (mmol/g)} = \frac{(V_{\text{DPP-THF}} - V_{\text{blank}}) \times 10^{-3} \times C_{\text{KOH}} \times 10^3}{W_{\text{DPP-THF}} \times 10^{-3}} \quad \text{S7}$$

Thermodynamic analysis. The thermodynamic data of oxidative cracking and thermal cracking of PP were calculated by Benson Group contribution method²⁷. The molar formation enthalpy and molar entropy of the substance were calculated as follows.

$$\Delta_f H_m^\theta(A) = \sum_k N_k \Delta_f H_{m,k}^\theta \quad \text{S8}$$

$$S_m^\theta(A) = \sum_k N_k S_{m,k}^\theta + S_{m,s}^\theta \quad \text{S9}$$

A represents the substance involved in the reaction, k represents the groups, and $S_{m,s}^\theta$ is the symmetric entropy of substance A.

The thermodynamic equations for the chemical reaction are as follows.

$$\Delta_r H_m^\theta = \sum_A v_A \Delta_f H_m^\theta(A) \quad \text{S10}$$

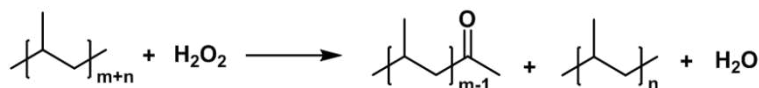
$$\Delta_r S_m^\theta = \sum_A v_A S_m^\theta(A) \quad \text{S11}$$

$$\Delta_r G_m^\theta = \Delta_r H_m^\theta - T \Delta_r S_m^\theta \quad \text{S12}$$

v_A is the stoichiometric number of the reaction.

The contribution values of the groups are in Extended Data Table 3. The chemical

equation for the oxidation cracking of PP to form methyl ketone is:



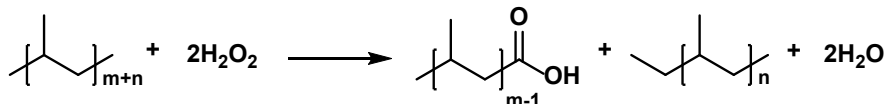
The total contribution of groups to this reaction is:

$$G_{r,m}(\text{oxidation craking}) = -2G_2 - G_3 + G_4 + G_5 + G_6 + G_{13} - G_{14} \quad \text{S13}$$

From this, the thermodynamic data for the oxidation cracking of PP were calculated

$$(\Delta_r H_{m,1}^\theta = -251.9 \text{ kJ/mol}, \Delta_r G_{m,1}^\theta = -296.2 \text{ kJ/mol}).$$

The contribution values of the groups are in Extended Data Table 3. The chemical equation for the oxidation cracking of PP to form carboxyl is:



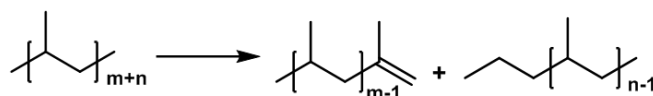
The total contribution of groups to this reaction is:

$$G_{r,m}(\text{oxidation craking}) = -G_2 - G_3 + G_5 + G_7 + G_8 + 2G_{13} - 2G_{14} \quad \text{S14}$$

From this, the thermodynamic data for the oxidation cracking of PP were calculated

$$(\Delta_r H_{m,2}^\theta = -594.8 \text{ kJ/mol}, \Delta_r G_{m,2}^\theta = -620.3 \text{ kJ/mol}).$$

The chemical equation for the thermal cracking of PP to form isopropenyl is:



The total contribution of groups to this reaction is:

$$G_{r,m}(\text{thermal craking}) = -G_1 - G_2 - 2G_3 + G_9 + G_{10} + G_{11} + G_{12} \quad \text{S15}$$

From this, the thermodynamic data for the thermal cracking of PP were calculated ($\Delta_r H_{m,3}^\theta =$

$$86.1 \text{ kJ/mol}, \Delta_r G_{m,3}^\theta = 36.9 \text{ kJ/mol}).$$

Products characterization

Gas chromatography. Methyl ester derivatized degradation solution and dimethyl

1 succinate standard were analyzed quantitatively on an Agilent 7890B gas chromatograph
2 equipped with a flame ionization detector (FID). Organic components in degradation solution
3 were analyzed qualitatively on a GCMS-QP2010 Plus gas Chromatograph/Mass
4 Spectrometer. Gas products were analyzed qualitatively on a Shimadzu GC-8AIT gas
5 chromatograph equipped with a thermal conductivity detector (TCD), using Ar as the internal
6 standard.

7 *Fourier-Transform Infrared Spectroscopy (FT-IR).* FT-IR spectroscopy was performed
8 on a Nicolet 6700 FT-IR System. Infrared testing was performed using potassium bromide
9 compressions with a wave number range of 4000-400 cm^{-1} .

10 *Nuclear Magnetic Resonance (NMR).* The degradation products were analyzed by a
11 Bruker Avance 400 Fouriertransform nuclear magnetic resonance spectrometer operating at
12 400 MHz (^1H NMR) or 600 MHz (^{13}C NMR). CDCl_3 was tested as a deuterated reagent.

13 *Gel permeation chromatography (GPC).* Molecular weight distributions of DPP-THF
14 were analyzed on an Agilent PL-GPC 220 gel permeation chromatograph, equipped with a
15 PL-Gel Mixed B guard column, three PL-Gel Mixed B columns, and a refractive index (RI)
16 detector. THF was used as a mobile phase. Molecular weight was determined from a
17 calibration curve obtained using polystyrene standards. The molecular weight of PP-T30S,
18 DPP-Xylene and commercial PP is obtained by high temperature GPC test, which were
19 analyzed on an Agilent PL-GPC 220 gel permeation chromatograph, equipped with PL-Gel
20 MIXED-B LS column, and a refractive index (RI) detector. 1,2,4-TCB (stabilized with BHT)
21 was used as a mobile phase at 150°C. Molecular weight was determined from a calibration
22 curve obtained using polystyrene standards.

23 *Total Organic Carbon.* The test of dissolved organic carbon in degradation solution was

1 performed on Analytik Jena MULTI N/C 2100S instrument.

Data availability

The authors declare that all data supporting the findings of this study are available in the article, its Extended Data, its Source Data or from the corresponding authors upon request.

References

- 1 Geyer, R., Jambeck, J. R. & Law, K. L. Production, use, and fate of all plastics ever made. *Sci. Adv.* **3** : e1700782 (2017).
- 2 Qiao, J. *et al.* Recent advances in polyolefin technology. *Polym. Chem.* **2**, 1611-1623 (2011).
- 3 Sharuddin, S. D. A., Abnisa, F., Daud, W. M. A. W. & Aroua, M. K. A review on pyrolysis of plastic wastes. *Energ. Convers. Manage.* **2016**, 308–326 (2016).
- 4 Mark, L. O., Cendejas, M. C. & Hermans, I. The Use of Heterogeneous Catalysis in the Chemical Valorization of Plastic Waste. *ChemSusChem* **13**, 5808–5836 (2020).
- 5 Liu, S., Kots, P. A., Vance, B. C., Danielson, A. & Vlachos, D. G. Plastic waste to fuels by hydrocracking at mild conditions. *Sci. Adv.* **7** : eabf8283 (2021).
- 6 Jia, X., Qin, C., Friedberger, T., Guan, Z. & Huang, Z. Efficient and selective degradation of polyethylenes into liquid fuels and waxes under mild conditions. *Sci. Adv.* **2** : e1501591 (2016).
- 7 Jehanno, C. *et al.* Critical advances and future opportunities in upcycling commodity polymers. *Nature* **603**, 803-814 (2022).
- 8 Fazekas, T. J. *et al.* Diversification of aliphatic C–H bonds in small molecules and polyolefins through radical chain transfer. *Science* **375**, 545–550 (2022).
- 9 Liye Chen *et al.* Selective, Catalytic Oxidations of C–H Bonds in Polyethylenes Produce Functional Materials with Enhanced Adhesion. *Chem* **7**, 137–145 (2021).
- 10 Elmanovich, I. V. *et al.* Thermal oxidation of polypropylene catalyzed by manganese oxide aerogel in oxygen-enriched supercritical carbon dioxide. *J. Supercrit. Fluid.* **158**, 104744 (2020).
- 11 Bäckström, E., Odelius, K. & Hakkarainen, M. Trash to Treasure: Microwave-Assisted Conversion of Polyethylene to Functional Chemicals. *Ind. Eng. Chem. Res.* **56**, 14814–14821 (2017).
- 12 Pifer, A. & Sen, A. Chemical Recycling of Plastics to Useful Organic Compounds by Oxidative Degradation. *Angew. Chem. Int. Ed.* **37**, 3306-3308 (1998).
- 13 Chow, C.-F., Wong, W.-L., Ho, K.-F., Chan, C.-S. & Gong, C.-B. Combined Chemical Activation and Fenton Degradation to Convert Waste Polyethylene into High-Value Fine Chemicals. *Chem. Eur. J.* **22**, 9513–9518 (2016).
- 14 Jiao, X. *et al.* Photocatalytic Conversion of Waste Plastics into C₂ Fuels under Simulated Natural Environment Conditions. *Angew. Chem. Int. Ed.* **59**, 15497–15501 (2020).
- 15 Chen, W.-T., Jin, K. & Wang, N.-H. L. Use of Supercritical Water for the Liquefaction of Polypropylene into Oil. *ACS Sustain. Chem. Eng.* **7**, 3749–3758 (2019).
- 16 Yao, D., Yang, H., Chen, H. & Williams, P. T. Investigation of nickel-impregnated zeolite catalysts for hydrogen/syngas production from the catalytic reforming of waste

- polyethylene. *Appl. Catal. B: Environ.* **227**, 477-487 (2018).
- 17 Jing, X., Yan, G., Zhao, Y., Wen, H. & Xu, Z. Study on mild cracking of polyolefins to liquid hydrocarbons in a closed batch reactor for subsequent olefin recovery. *Polym. Degrad. Stabil.* **109**, 79-91 (2014).
- 18 Celik, G. *et al.* Upcycling Single-Use Polyethylene into High-Quality Liquid Products. *ACS Cent. Sci.* **5**, 1795–1803 (2019).
- 19 Rorrer, J. E., Troyano-Valls, C., Beckham, G. T. & Román-Leshkov, Y. Hydrogenolysis of Polypropylene and Mixed Polyolefin Plastic Waste over Ru/C to Produce Liquid Alkanes. *ACS Sustain. Chem. Eng.* **9**, 11661–11666 (2021).
- 20 Jia, C. *et al.* Deconstruction of high-density polyethylene into liquid hydrocarbon fuels and lubricants by hydrogenolysis over Ru catalyst. *Chem Catal.* **1**, 437–455 (2021).
- 21 Tennakoon, A. *et al.* Catalytic upcycling of high-density polyethylene via a processive mechanism. *Nat. Catal.* **3**, 893-901 (2020).
- 22 Zhang, F. *et al.* Polyethylene upcycling to long-chain alkylaromatics by tandem hydrogenolysis/aromatization. *Science* **370**, 437–441 (2020).
- 23 Kanbur, U. *et al.* Catalytic carbon–carbon bond cleavage and carbon-element bond formation give new life for polyolefins as biodegradable surfactants. *Chem* **7**, 1–16 (2021).
- 24 Salmela, M., Lehtinen, T., Efimova, E., Santala, S. & Santala, V. Towards bioproduction of poly- α -olefins from lignocellulose. *Green Chem.* **22**, 5067–5076 (2020).
- 25 Jiang, H. *et al.* Synthesis of Long-Subchain Hyperbranched Polypropylene Using Thermally Degraded Products as Precursor. *Macromolecules* **54**, 5567–5576 (2021).
- 26 Häußler, M., Eck, M., Rothauer, D. & Mecking, S. Closed-loop recycling of polyethylene-like materials. *Nature* **590**, 423-427 (2021).
- 27 Pitet, L. M. & Hillmyer, M. A. Carboxy-Telechelic Polyolefins by ROMP Using Maleic Acid as a Chain Transfer Agent. *Macromolecules* **44**, 2378–2381 (2011).
- 28 Witt, T., Stempfle, F., Roesle, P., Häußler, M. & Mecking, S. Unsymmetrical α,ω -Difunctionalized Long-Chain Compounds via Full Molecular Incorporation of Fatty Acids. *ACS Catal.* **5**, 4519–4529 (2015).
- 29 Zibek, S., Huf, S., Wagner, W., Hirth, T. & Rupp, S. Fermentative Herstellung der α,x -Dicarbonsäure 1,18-Oktadecendisäure als Grundbaustein für biobasierte Kunststoffe. *Chem. Ing. Tech.* **81**, 1797-1808 (2009).
- 30 Sawaguchit, T. & Seno, M. Preparation of Telechelic Oligomers by the Thermal Degradation of Syndiotactic Polypropylene. *Polym. J.* **28**, 817-820 (1996).
- 31 Zeng, M. *et al.* Chemical Upcycling of Polyethylene to Value-Added α,ω -DivinylFunctionalized Oligomers. *ACS Sustain. Chem. Eng.* **9**, 13926–13936 (2021).
- 32 Chow, C.-F., Wong, W.-L., Chan, C.-W. & Chan, C.-S. Converting inert plastic waste into energetic materials: A study on the light-accelerated decomposition of plastic waste with the Fenton reaction. *Waste Manage.* **75**, 174–180 (2018).
- 33 Girois, S., Audouin, L., Delprat, P. & Verdu, J. Weight loss mechanism in the photooxidation of polypropylene. *Polym. Degrad. Stabil.* **51**, 133-141 (1996).
- 34 Lacoste, J., Vaillant, D. & Carlsson, D. J. Gamma - , photo - , and thermally - initiated oxidation of isotactic polypropylene. *J. Polym. Sci. Pol. Chem.* **31**, 715-722 (1993).

Acknowledgements

This work is supported by the National Natural Science Foundation of China (No. 51721091), the 111 project (B20001), the Fundamental Research Funds for the Central Universities (2020SCUNL205) and State Key Laboratory of Polymer Materials Engineering (No.sklpme2020-1-02).

Author contributions

S. X. and X. W. conceived the concept. X. W. designed and performed the experiments and wrote the manuscript. All authors analyzed the experiments. S. X., F. Z. and Y. W. provided theoretical guidance and edited the manuscript.

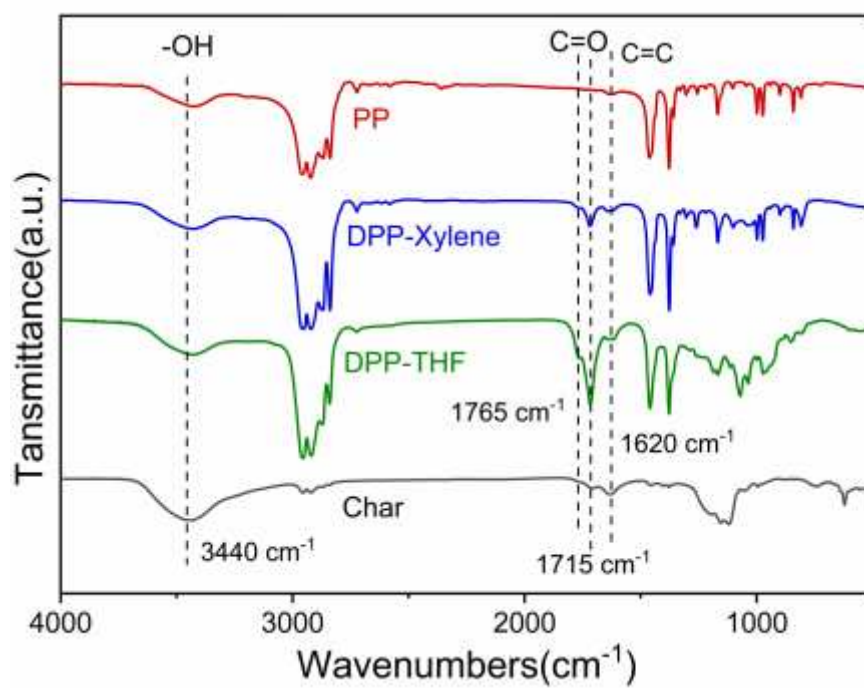
Competing interests

The authors declare that they have no competing interests.

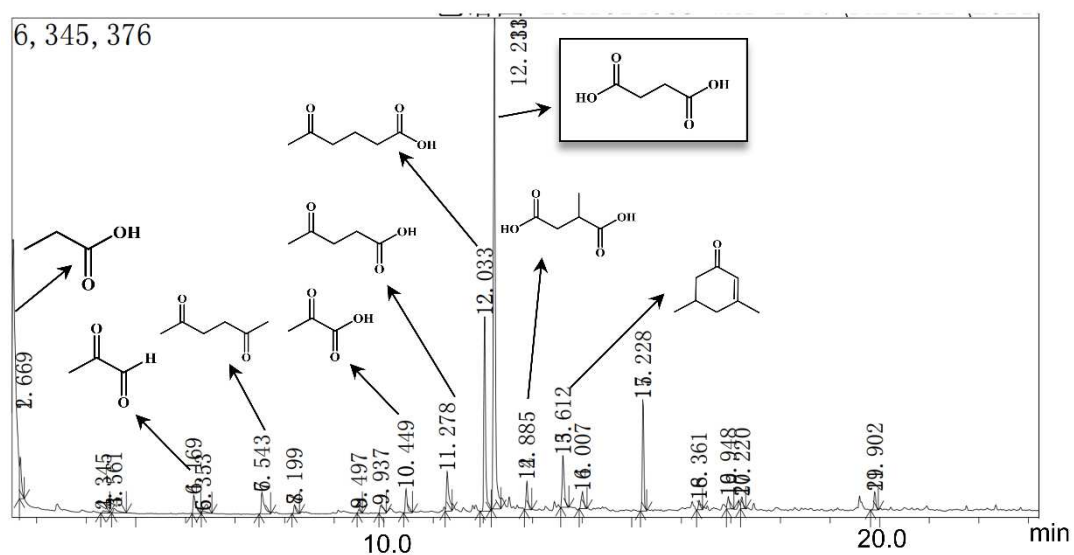
Materials & Correspondence.

Correspondence and requests for materials should be addressed to Yu-Zhong Wang.

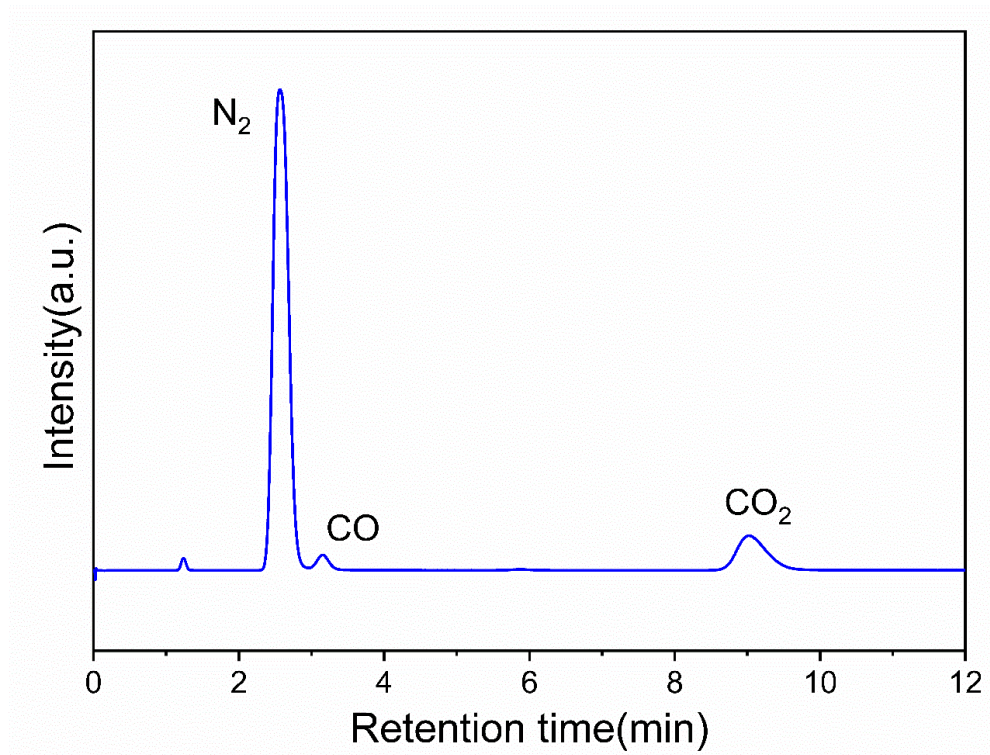
Extended data



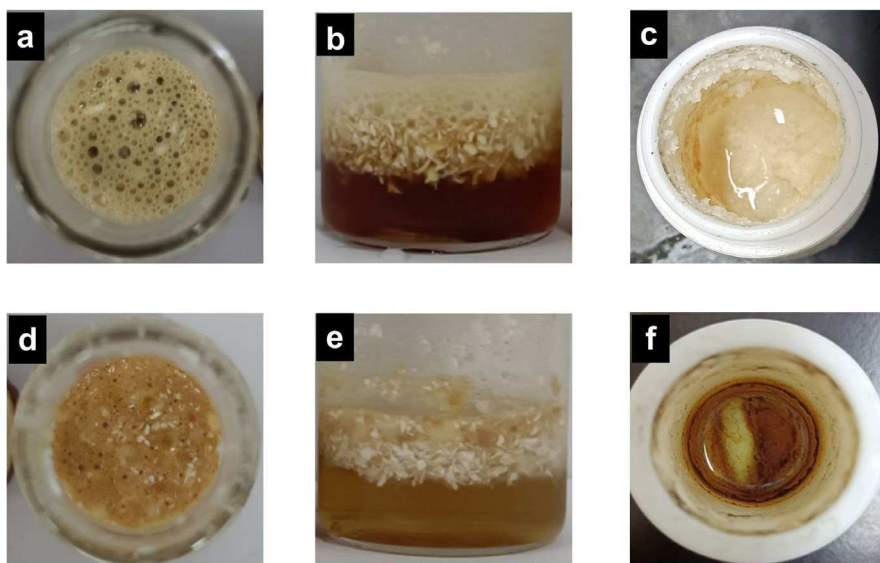
Extended Data Fig. 1 | FT-IR spectra of PP, DPP-Xylene, DPP-THF and Char.



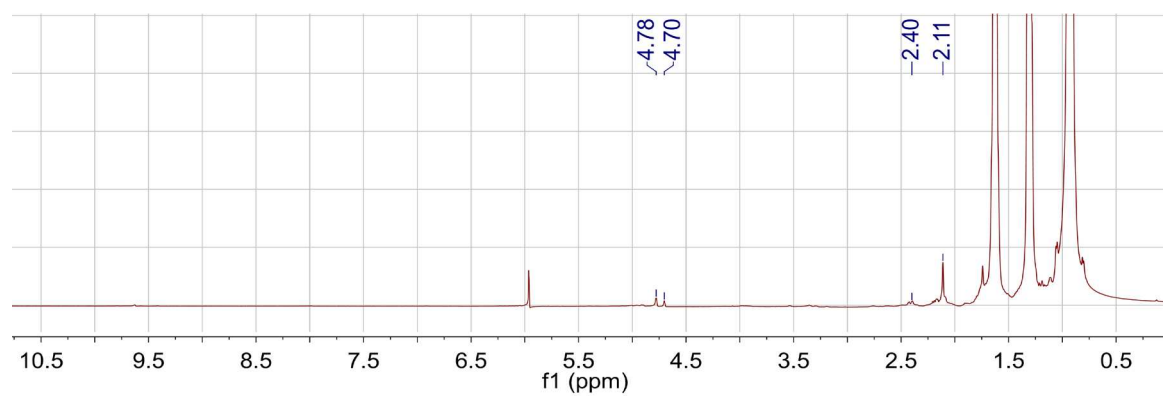
Extended Data Fig. 2 | GC-MS analysis of degradation solution by dichloromethane (DCM) extraction.



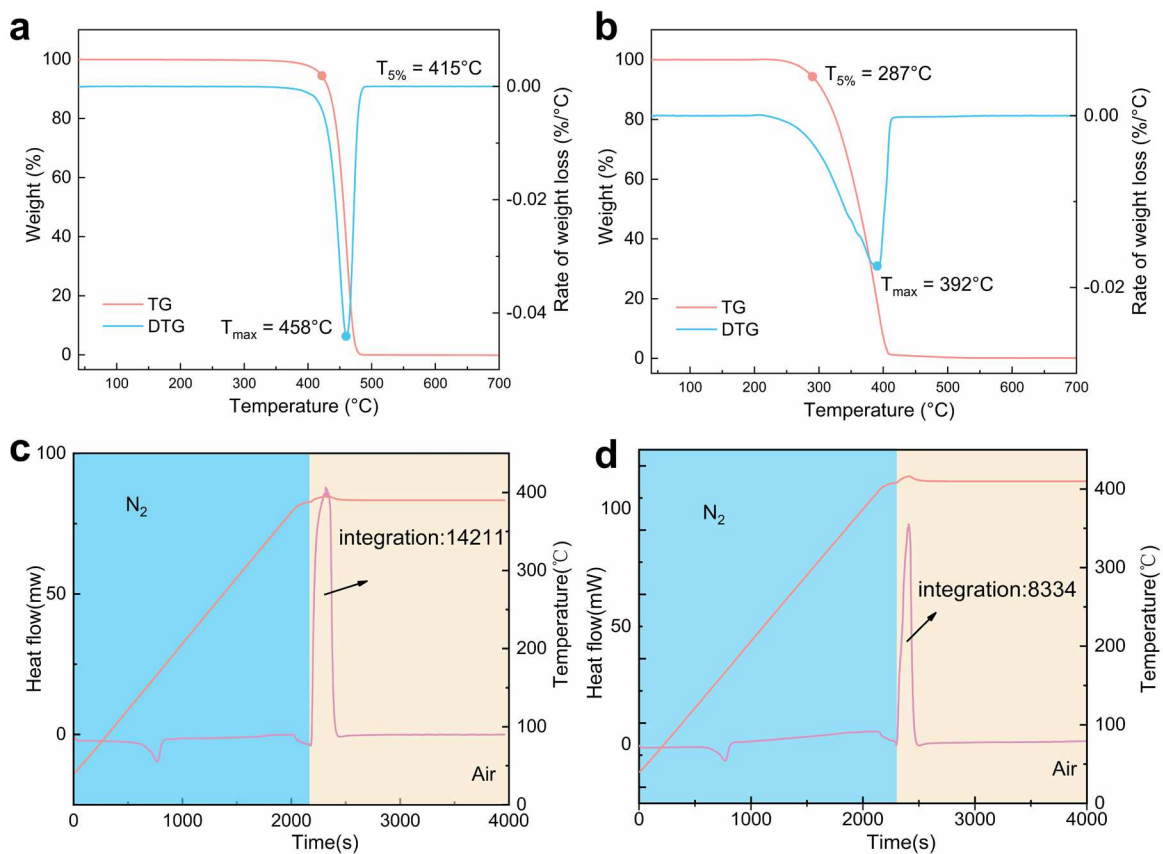
Extended Data Fig. 3 | GC-TCD of gases collected from the degradation of PP at 80°C for 30 min.



Extended Data Fig. 4 | The adsorption of PP to iron ions in H_2O_2 solution containing CTAB: top **(a)**, front **(b)**; The degradation of PP with the assistance of CTAB **(c)**. The adsorption of PP to iron ions in H_2O_2 solution containing SDS: top **(d)**, front **(e)**; The degradation of PP with the assistance of SDS **(f)**.

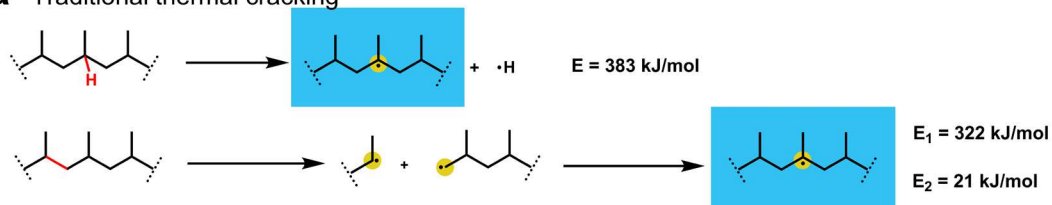


Extended Data Fig. 5 | 400 MHz ¹H NMR spectra of DPP-Xylene in 1,1,2,2-C₂D₂Cl₄ at 120°C.

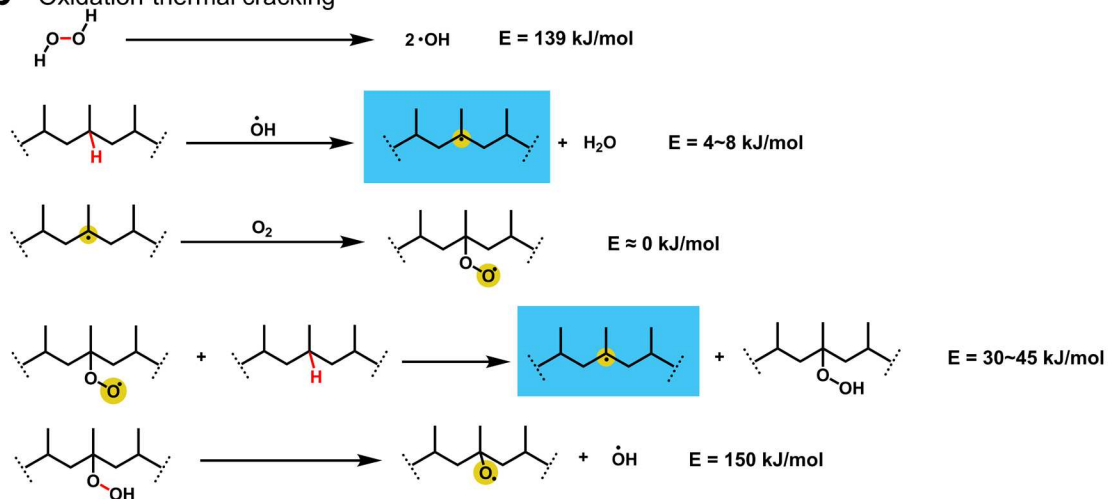


Extended Data Fig. 6 | Decomposition of PP from 40°C to 700°C under (a) N_2 atmosphere; (b) Air atmosphere. DSC of thermostatic decomposition of PP under air atmosphere at (c) 390°C ; (d) 410°C .

a Traditional thermal cracking



b Oxidation-thermal cracking



1

2

Extended Data Fig. 7 | Generation of tertiary carbon-alkyl radicals by **(a)** traditional thermal

3

cracking; **(b)** oxidative-thermal cracking.

Extended Data Table 1 | PP was converted to DPP-THF and DPP-Xylene at different ratio of SDS to PP.

Entry	SDS:PP	DPP-THF					
		Yield (wt%)	Mw (10 ³ Da)	PDI	Methyl ketone * (mmol/g)	Isopropenyl * (mmol/g)	Carboxyl † (mmol/g)
1	0.125	tace	-	-	-	-	-
2	0.30	53.5	1.97	1.94	0.92	0.40	1.42
3	0.35	58.8	1.79	1.89	1.03	0.40	1.02
4	0.40	60.0	1.56	1.95	1.29	0.56	1.26
5	0.45	57.2	1.74	2.00	1.11	0.50	1.42
6	0.50	50.5	1.78	1.74	0.80	0.50	1.20
7	0.875	23.1	2.02	1.97	0.68	0.17	0.81

*Calculated from ¹H NMR, using an internal standard. † Calculated by titration experiment using KOH ethanol solution.

1 **Extended Data Table 2 | Oxygen contents in DPP-THF.**

Entry	SDS:PP	Methyl ketone (mmol/g)	Carboxyl (mmol/g)	O _{C=O} % *	O _{COOH} % *	O _{C=O} % + O _{COOH} %	O _{EA} % †
1	0.30	0.92	1.42	1.47	4.54	6.02	6.63
2	0.35	1.03	1.02	1.65	3.26	4.91	6.63
3	0.40	1.29	1.26	2.07	4.03	6.10	6.51
4	0.45	1.11	1.42	1.78	4.54	6.32	7.42
5	0.50	0.80	1.20	1.27	3.84	5.11	7.15
6	0.875	0.68	0.81	1.09	2.59	3.68	3.61

2

3 * Calculated according to the corresponding group content. † Obtained by organic element analysis (EA).

Extended Data Table 3 | Benson group increments for oxidation cracking and thermal cracking of PP.

Group		$\Delta_f H_m^\theta$ (kJ/mol)	S_m^θ (J/mol)	Number of contributions to reaction		
				Methyl ketone	Carboxyl	Isopropenyl
G ₁	CH ₃ -(C)	-42.19	127.29	0	0	-1
G ₂	CH ₂ -(2C)	-20.64	39.43	-2	-1	-1
G ₃	CH-(3C)	-7.95	-50.52	-1	-1	-2
G ₄	CH ₃ -(CO)	-42.19	127.29	1	0	0
G ₅	CH ₂ -(C,CO)	-21.77	40.18	1	1	0
G ₆	CO-(2C)	-131.44	62.83	1	0	0
G ₇	CO-(C,O)	-146.96	20.01	0	1	0
G ₈	OH-(CO)	-243.25	102.66	0	1	0
G ₉	CH ₃ -(=C)	-42.19	127.29	0	0	1
G ₁₀	CH ₂ -(C,=C)	-19.92	41.02	0	0	1
G ₁₁	=CH ₂	26.2	115.57	0	0	1
G ₁₂	=C-(2C)	43.28	-53.16	0	0	1
G ₁₃	H ₂ O	-241.82	188.7	1	2	0
G ₁₄	H ₂ O ₂	-136.1	232.9	-1	-2	0